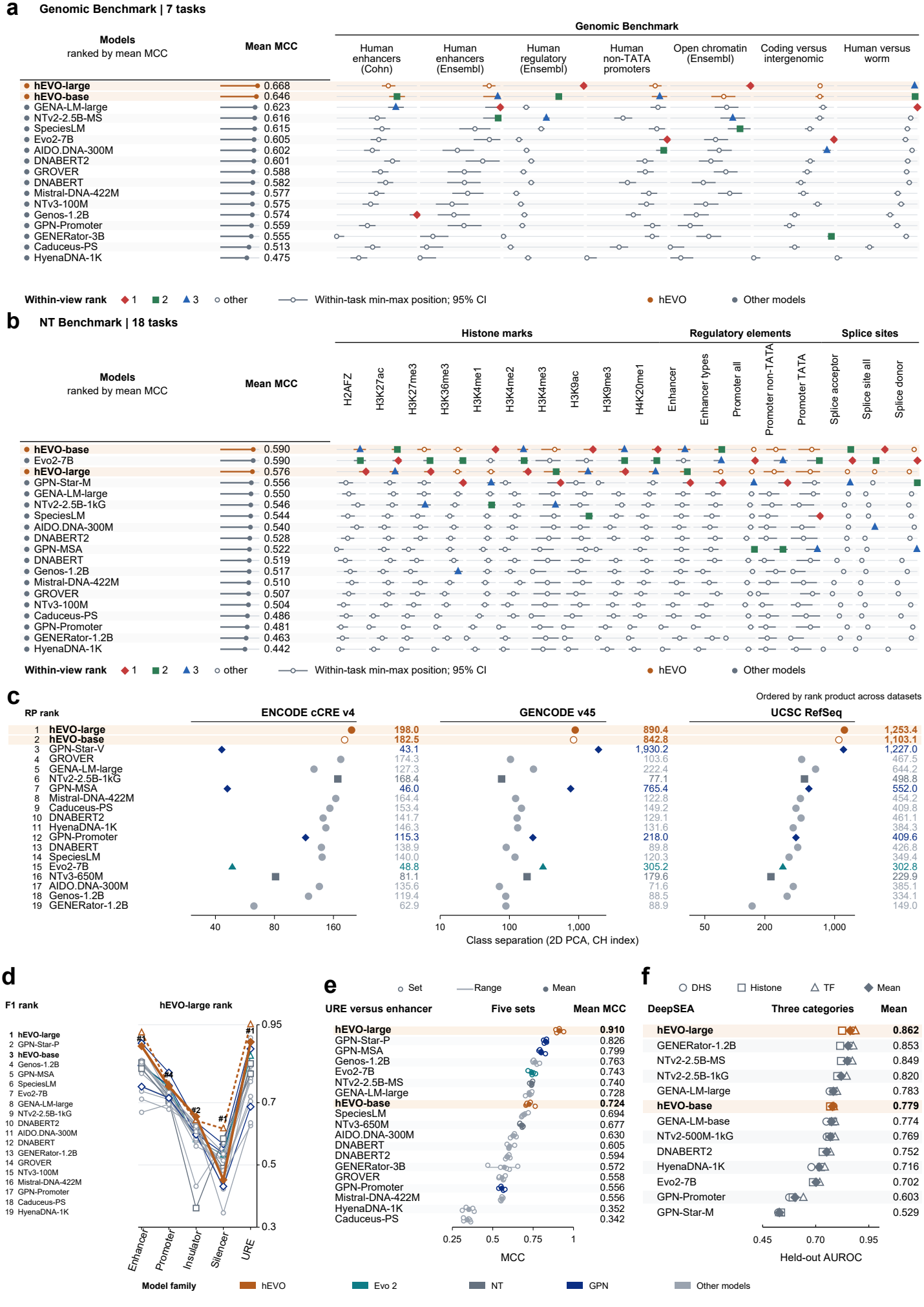

a
Genomic Benchmark | 7 tasks
Models
ranked by mean MCC
Mean MCC
Genomic Benchmark
Human enhancers (Cohn)
Human enhancers (Ensembl)
Human regulatory (Ensembl)
Human non-TATA promoters
Open chromatin (Ensembl)
Coding versus intergenomic
Human versus worm
hEVO-large 0.668
hEVO-base 0.646
GENA-LM-large 0.623
NTv2-2.5B-MS 0.616
SpeciesLM 0.615
Evo2-7B 0.605
AIDO.DNA-300M 0.602
DNABERT2 0.601
GROVER 0.588
DNABERT 0.582
Mistral-DNA-422M 0.577
NTv3-100M 0.575
Genos-1.2B 0.574
GPN-Promoter 0.559
GENERator-3B 0.555
Caduceus-PS 0.513
HyenaDNA-1K 0.475
Within-view rank 1 2 3 other
Within-task min-max position; 95% CI
hEVO
Other models
b
NT Benchmark | 18 tasks
Models
ranked by mean MCC
Mean MCC
Histone marks
Regulatory elements
Splice sites
H2AFZ
H3K27ac
H3K27me3
H3K36me3
H3K4me1
H3K4me2
H3K4me3
H3K9ac
H3K9me3
H4K20me1
Enhancer
Enhancer types
Promoter all
Promoter non-TATA
Promoter TATA
Splice acceptor
Splice site all
Splice donor
hEVO-base 0.590
Evo2-7B 0.590
hEVO-large 0.576
GPN-Star-M 0.556
GENA-LM-large 0.550
NTv2-2.5B-1kG 0.546
SpeciesLM 0.544
AIDO.DNA-300M 0.540
DNABERT2 0.528
GPN-MSA 0.522
DNABERT 0.519
Genos-1.2B 0.517
Mistral-DNA-422M 0.510
GROVER 0.507
NTv3-100M 0.504
Caduceus-PS 0.486
GPN-Promoter 0.481
GENERator-1.2B 0.463
HyenaDNA-1K 0.442
Within-view rank 1 2 3 other
Within-task min-max position; 95% CI
hEVO
Other models
c
Ordered by rank product across datasets
RP rank
ENCODE cCRE v4
GENCODE v45
UCSC RefSeq
1 hEVO-large 198.0 890.4 1,253.4
2 hEVO-base 182.5 842.8 1,103.1
3 GPN-Star-V 43.1 1,930.2 1,227.0
4 GROVER 174.3 103.6 467.5
5 GENA-LM-large 127.3 222.4 644.2
6 NTv2-2.5B-1kG 168.4 77.1 498.8
7 GPN-MSA 46.0 765.4 552.0
8 Mistral-DNA-422M 164.4 122.8 454.2
9 Caduceus-PS 153.4 149.2 409.8
10 DNABERT2 141.7 129.1 461.1
11 HyenaDNA-1K 146.3 131.6 384.3
12 GPN-Promoter 115.3 218.0 409.6
13 DNABERT 138.9 89.8 426.8
14 SpeciesLM 140.0 120.3 349.4
15 Evo2-7B 48.8 305.2 302.8
16 NTv3-650M 81.1 179.6 229.9
17 AIDO.DNA-300M 135.6 71.6 385.1
18 Genos-1.2B 119.4 88.5 334.1
19 GENERator-1.2B 62.9 88.9 149.0
40 80 160
10 100 1,000
50 200 1,000
Class separation (2D PCA, CH index)
d
F1 rank
hEVO-large rank
1 hEVO-large
2 GPN-Star-P
3 hEVO-base
4 Genos-1.2B
5 GPN-MSA
6 SpeciesLM
7 Evo2-7B
8 GENA-LM-large
9 NTv2-2.5B-1kG
10 DNABERT2
11 AIDO.DNA-300M
12 DNABERT
13 GENERator-1.2B
14 GROVER
15 NTv3-100M
16 Mistral-DNA-422M
17 GPN-Promoter
18 Caduceus-PS
19 HyenaDNA-1K
0.95
0.7
0.5
0.3
Enhancer
Promoter
Insulator
Silencer
URE
Model family
hEVO
Evo 2
NT
GPN
Other models
e
Set
Range
Mean
URE versus enhancer
Five sets
Mean MCC
hEVO-large 0.910
GPN-Star-P 0.826
GPN-MSA 0.799
Genos-1.2B 0.763
Evo2-7B 0.743
NTv2-2.5B-MS 0.740
GENA-LM-large 0.728
hEVO-base 0.724
SpeciesLM 0.694
NTv3-650M 0.677
AIDO.DNA-300M 0.630
DNABERT 0.605
DNABERT2 0.594
GENERator-3B 0.572
GROVER 0.558
GPN-Promoter 0.556
Mistral-DNA-422M 0.556
HyenaDNA-1K 0.352
Caduceus-PS 0.342
0.25 0.5 0.75 1
MCC
f
DHS
Histone
TF
Mean
DeepSEA
Three categories
Mean
hEVO-large 0.862
GENERator-1.2B 0.853
NTv2-2.5B-MS 0.849
NTv2-2.5B-1kG 0.820
GENA-LM-large 0.783
hEVO-base 0.779
GENA-LM-base 0.774
NTv2-500M-1kG 0.769
DNABERT2 0.752
HyenaDNA-1K 0.716
Evo2-7B 0.702
GPN-Promoter 0.603
GPN-Star-M 0.529
0.45 0.70 0.95
Held-out AUROC